

Braden K. Webb

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WORK EXPERIENCE

Meta Reality Labs, contracted through Cognara

Remote from Boston, MA

AI Software Engineer (*contingent worker*)

May 2025 - Present

- Develop agentic AI platform to automate manual bug reproduction & test validation processes for engineers and QA leads, scaling the tool from 0 to 300+ weekly active (internal) users
- Write device-control MCP server to enable agentic interaction w/mobile phones, VR headsets, & wearables

Pacific Northwest National Laboratory

Remote from Boston, MA

ML Research Associate - AI & Data Analytics Division

May 2023 - May 2025

- Pattern-of-life modeling and algorithmic anomaly detection through graph analytics for national security threat prevention on the Sigma+ program; sponsors include DARPA and other government agencies
- Built full-stack web app with TypeScript, React, and Next.js, leveraging Elasticsearch, Postgres, and Apache NiFi for data flow, deploying with AWS, and contributing to renewal of \$5M contract
- Implemented RAG pipeline & trained LORA adapters to automate mappings between related taxonomies of threat patterns for cyber-physical systems
- Published and empirically validated formal mathematical model to maximize security of cyber-physical systems

BYU Computer Science Department

Provo, UT

Machine Translation Research Assistant @ MATRIX Labs

Sep. 2022 - May 2023

- Created data cleaning pipeline for dynamically processing and aligning translation memory files for 90+ languages, preparing corpus for publication in Linguistic Data Consortium
- Harnessed distributed HPC GPUs to train 800M-parameter, 24-language neural machine translation model from scratch to investigate effects of language family tags on translation quality

NLP Research Assistant @ DRAGN Labs

Apr. 2022 - Jan. 2023

- Fine-tuned GPT-2 for quest and dialogue generation with domain-specific text obtained through web-scraping
- Developed novel metric for evaluating relevance of procedurally generated NPC dialogue to quest objectives, and validated its correlation to human preferences via surveys and user studies, leading to an academic publication

Computational Research Assistant

Dec. 2020 - Apr. 2022

- Evaluated open-source tools for English to IPA transliteration, detected and traced surname transcription errors in historic census data, and processed millions of records using Pandas, NumPy, and Epitran

Qualtrics International, Inc

Remote from Provo, UT

Single Sign-On Technology Consultant

June 2021 - Apr. 2022

- Provided support integrating SAML, CAS, LDAP, and Google OAuth SSO protocols for 50+ clients

EDUCATION

B.S. Mathematics | Brigham Young University

Provo, UT

Minors: Computer Science, Linguistics, & Logic

April 2023

- GPA: 3.9/4.0, 4-Year Full-Tuition Academic Scholarship, National Merit Scholarship
- Coursework including deep learning, predictive modeling, natural language processing, algorithm design & analysis

PUBLICATIONS

- [1] A. Shurtz, L. Sorenson, **B. Webb**, et al. *MMMC: A Massively Multi-way-aligned Multilingual Corpus*. In progress.
- [2] **B. Webb**, R. Meyur, S. Purohit. Submitted to *IEEE Big Data '24*. [Link](#)
- [3] R. Meyur, S. Purohit, **B. Webb**. *AAAI-24 Workshop on AICS*, Feb 26, 2024, Vancouver, Canada. [Link](#)
- [4] T. Ashby, **B. Webb**, et al. *CHI '23*, April 23-28, 2023, Hamburg, Germany. [Link](#)

SKILLS

Computer Languages: Python, TypeScript, R (limited), C++, Hack, Java, SQL

ML Frameworks: PyTorch, JAX, NumPy, Pandas, SciPy, spaCy, scikit-learn, Hugging Face, OpenNMT, NLTK, Neo4j

Other Tools: Docker, Bash, GraphQL, Git, Android, Azure AI, QGIS, Excel, Slurm, SSO, L^AT_EX, AWS Sagemaker

Foreign Languages: Slovak (proficient), Czech (proficient), Russian (basic reading ability)